25-11-2025

INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

**End Semester Examination, November - December, 2025**

Degree: **B.Tech** Semester: **First**

Stream: **BT / CSE(AI)/ CSE(AI & ML)/ CSE(Cybersecurity)/ CSE (IOT, CS, BT)/ CSE(Networks)/ CSBS/ CSE/ ECE/ EEE/ EE/ IT/ ME**

Paper Code: **ESCEC101**

Paper Name: **Basic Electronics Engineering**

Full Marks: **100** Duration: **3 Hours**

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Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1. What is the role of doping in semiconductors? 2
2. Define active components. 2
3. What is the role of temperature in carrier concentration? 2
4. State the condition for forward bias in a diode. 2
5. What is the efficiency of a half-wave rectifier? 2
6. Define a Bipolar Junction Transistor (BJT). 2
7. Sketch the V-I characteristics of a BJT in CE mode. 2
8. What is the function of a BJT as a switch? 2
9. What is the role of the gate in a MOSFET? 2
10. What is the use of MOSFET as an amplifier? 2
11. What is the main application of IGBT? 2
12. Why is high CMRR important in Op-Amp circuits? 2

13. Name two applications of differential amplifiers. 2

14. What is TTL? State the applications of TTL. 2

15. Why are NAND and NOR called universal gates? 2

**Part - B**

**Attempt any 6 questions out of 9 questions**

**Each question carries 5 Marks (5 × 6)**

16. Calculate diffusion current for $D=25cm^2/s$, $dn/dx=10^{20}\ cm^{-4}$. 5

17. Compare active and passive components with examples. 5

18. Design a voltage regulator using Zener diode. 5

19. A *npn* transistor with $\alpha=0.98$ is operated in CB mode. If the emitter current is 3 mA and reverse saturation current $I_{co} = 10\mu A$, what are the base and collector current? 5

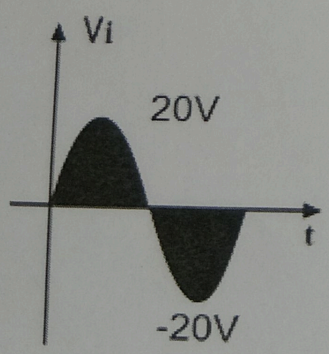
20. Explain the structure and working of an IGBT. 5

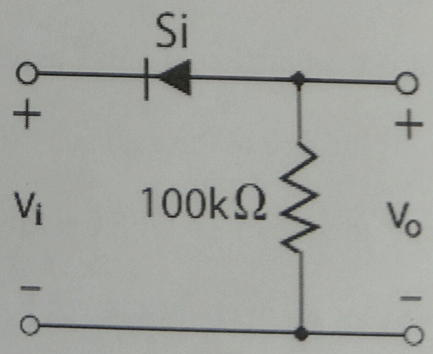
21. Explain how a MOSFET acts as a switch. 5

22. Draw the circuit diagram of an inverting amplifier using an Op-Amp. Derive the expression for voltage gain. Mention two advantages and one limitation of this configuration. 5

23. Show how NAND gates can be used to implement the following gates: NOT, AND, and OR Provide logic diagrams. 5

24. Explain and draw the output waveform pattern for the clipper circuit given below. 5



## Part - C

**Attempt any 5 questions out of 8 questions**

**Each question carries 10 Marks (10 × 5)**

25. For silicon at 300K, calculate intrinsic carrier concentration using $E_g = 1.1eV$. **10**

26. With circuit diagram, explain forward bias and reverse bias operations of a semiconductor junction diode. **10**

27. What is a Clamper Circuit? Design a clamper circuit to shift a sine wave signal to a required DC level and explain its working. **10**

28. Derive the expression for collector current using minority carrier distribution. **10**

29. Design a transistor switch circuit to control a 12V relay using a 5V logic signal. **10**

30. Compare planar MOSFETs with FinFETs in terms of structure, performance, and scaling. **10**

31. Explain positive and negative feedback with the help of Op-Amp circuits. Describe the advantages and disadvantages of each type and give one example circuit for both cases. **10**

32. Explain the working of XOR and XNOR gates using truth tables, symbols, and Boolean expressions. Discuss their importance in arithmetic circuits like adders and parity checkers with examples. **10**

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